ASSIGNMENT – 2

Foundations of Cryptography | BCA | Course: BCA309T

Q1. List the four fundamental cryptographic goals/objectives.

The four main objectives are confidentiality, integrity, authentication, and non-repudiation.

Q2. Define confidentiality with one example.

Confidentiality means only authorized people can read information. For example, encryption protects messages so outsiders cannot read them.

Q3. Explain confidentiality, integrity, authentication, and non-repudiation with examples. Why are all important?

Confidentiality keeps data secret, such as an encrypted banking transaction. Integrity ensures data is not changed without detection, such as checking a message authentication code. Authentication verifies who is communicating, such as a website certificate. Non-repudiation provides evidence of an action, such as a digital signature on a document. Together, these objectives help keep communication private, accurate, trustworthy, and accountable.

Q4. Difference between substitution and transposition ciphers, with examples.

A substitution cipher replaces each character with another character; for example, a Caesar cipher shifts letters. A transposition cipher rearranges the positions of characters without changing the characters themselves; for example, a rail fence cipher.

Q5. Discuss the evolution from classical ciphers to DES, AES, RSA, and SHA.

Classical ciphers used manual letter substitution or rearrangement and were vulnerable to pattern analysis. DES brought computer-based symmetric encryption but its key became too short against modern computing power. AES replaced DES as a stronger standard for protecting data. RSA introduced public-key cryptography, supporting secure key exchange and digital signatures. SHA hash functions help check data integrity; they are not encryption algorithms. These changes were driven by stronger computers and growing security needs.

Q6. What is a cryptographic key, and why is key management important?

A cryptographic key is a value used by an algorithm to encrypt, decrypt, or sign data. Key management is important because keys must be generated, stored, shared, and replaced securely; a stolen key can compromise protected information.

Q7. Explain how cryptography secures computer networks.

Encryption, such as TLS encryption, protects data while it travels across a network. Authentication using certificates helps verify that a server is genuine and can prevent unauthorized connections. Integrity checks, such as message authentication codes or digital signatures, help detect tampering. For example, when a user logs in to a banking website, TLS helps protect the session from eavesdropping and alteration.

Q8. Difference between passive and active attacks, with examples.

A passive attack secretly observes communication without changing it, such as eavesdropping on network traffic. An active attack changes, injects, or disrupts data, such as modifying a message during transmission.

Q9. How did cryptography evolve to counter threats, and how did public-key cryptography solve key distribution?

Classical ciphers were replaced by stronger mechanical and computer-based methods as codebreaking improved. Symmetric systems use one shared secret key, which creates a challenge: the sender and receiver must share it securely first. Public-key cryptography introduced a public key that can be shared openly and a private key kept secret. This helps with key exchange and digital signatures, reducing the need to send a shared secret directly. Modern systems commonly use public-key methods to establish a session key, then symmetric encryption to protect large amounts of data efficiently.

Q10. Give one modern symmetric and one modern asymmetric algorithm.

AES is a modern symmetric encryption algorithm. RSA is a widely used asymmetric algorithm.

Q11. Classify classical and modern cryptography and explain the transition.

Classical cryptography used manual or mechanical techniques, such as Caesar and Vigenère ciphers, mainly to substitute or rearrange letters. Modern cryptography uses mathematical and computer-based algorithms. Important milestones include the Enigma machine during World War II, DES for computer-era encryption, RSA and public-key cryptography in the 1970s, AES as a stronger standard, and modern hash functions such as SHA. Modern systems combine encryption, authentication, and integrity checks to protect digital communication.